

7 Iron

Iron is needed for chlorophyll formation and energy reactions. Many soils contain iron, yet roots cannot obtain enough when pH, bicarbonate, oxygen, or root health restrict availability.

Why it matters

- **Green tissue and energy.** Iron is required for chlorophyll formation and for electron transfer in photosynthesis and respiration.
- **Immobile in the plant.** Shortage symptoms usually appear first in young leaves because iron is not readily moved from older tissue.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists about 20 to 40 ppm as deficient and 50 to 200 ppm as adequate; it does not give a simple excess cutoff.
Leaf-test caution	Leaf iron can appear adequate even when new leaves show chlorosis. Surface contamination can also inflate the result.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH and carbonates	High pH, lime, and bicarbonate favor poorly soluble iron forms and can interfere with root iron use.
Microbes and chelates	Roots and microbes can produce organic acids and metal-binding compounds. Commercial chelate performance depends on the ligand, pH, placement, water, and root condition.
Organic matter	Organic compounds can bind and transport iron, but decomposition in saturated soil also consumes oxygen and changes redox chemistry.
Irrigation and aeration	Cold, wet, compacted, or waterlogged soil weakens roots. Bicarbonate-rich irrigation water can sustain chlorosis even when total soil Fe is high.
Root disease	Damaged feeder roots reduce uptake and can make an iron treatment appear ineffective.

Field clues and checks

- **Look at young leaves.** Iron chlorosis typically begins between veins while veins remain greener; severe leaves may become nearly pale yellow.
- **Test the cause.** Check soil and water pH, bicarbonate, lime, drainage, aeration, salinity, and feeder-root health.
- **Match chelate to chemistry.** If a chelate is justified, confirm that it remains stable at the measured pH and evaluate new growth against an untreated comparison.

DST interpretation An iron signal should open a pH, bicarbonate, irrigation, and root-health investigation. Soil iron quantity or one leaf number alone is not enough to choose a treatment.